

Intro to Lasers

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Objectives

- Learn basic Inkscape design
- Importing a design for cutting
- Learn to operate under supervision by an authorized person
- Safe Operation



Requirements

- Must be supervised by authorized user
- Must use only during office hours or open-house.
- \$20 Donation to Spark

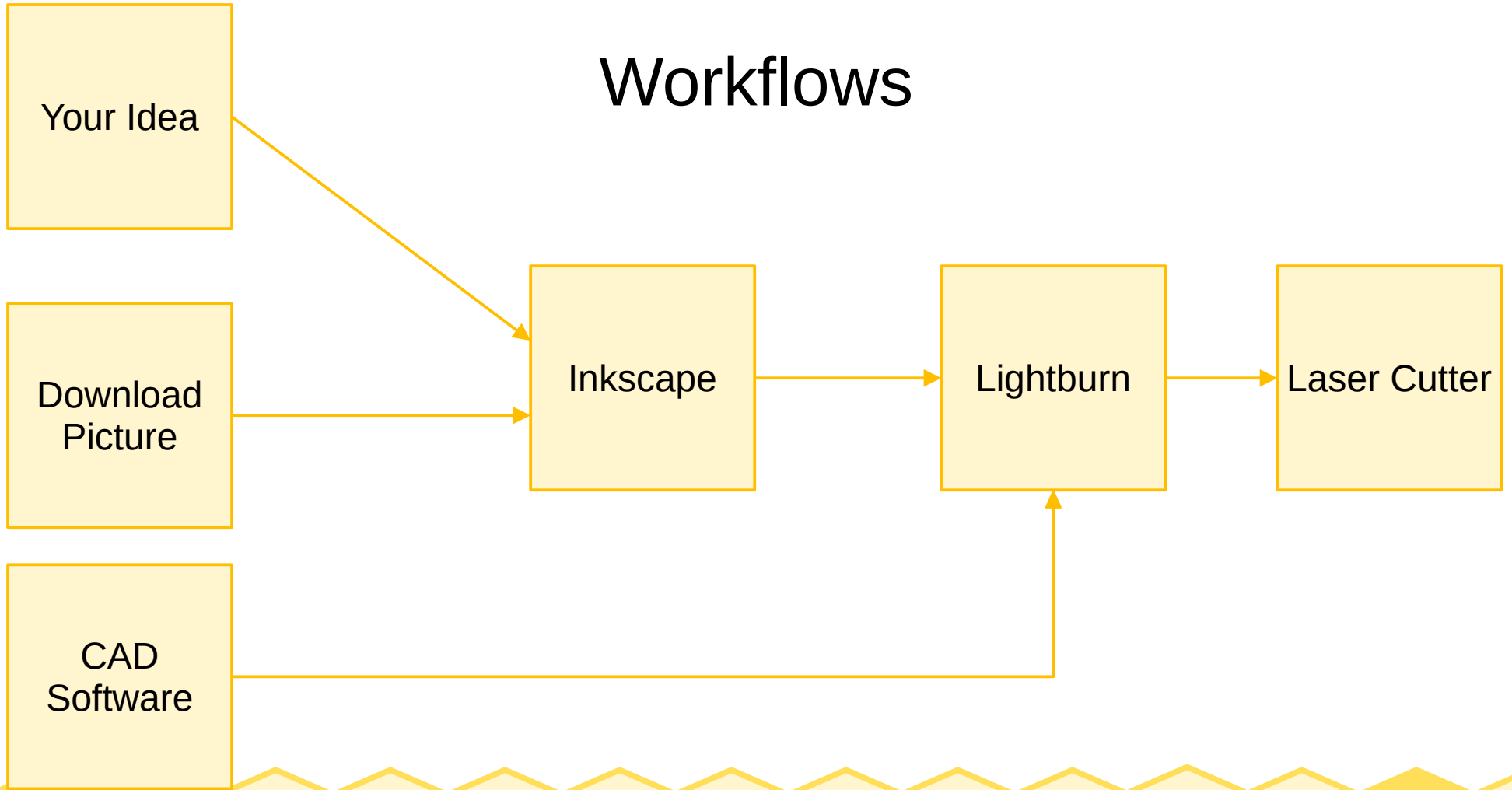


Topics Covered

- We will go through a basic design in Inkscape
- We will import file into Lightburn for cutting
- We will set up cutter for operation
- You will laser etch your own cutting board to take home



Workflows



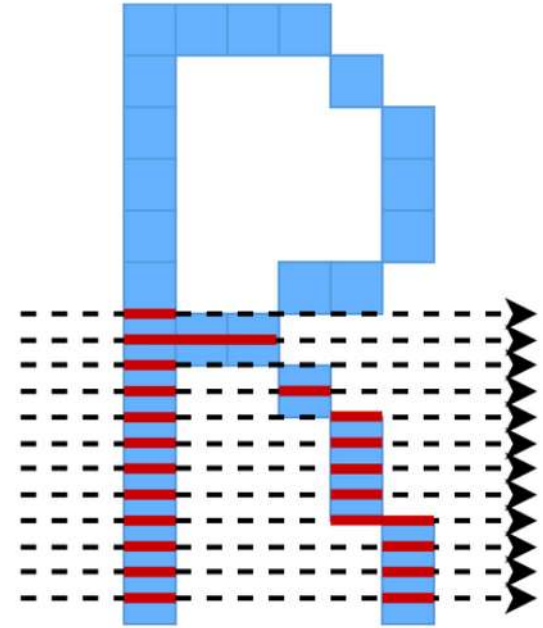
Vector and Raster images

- Vectors are paths that the laser will follow
-
- Raster images are “pixels” or points that will be etched



Cutting raster images

- Raster images will be converted to black and white
- Laser head will “Scan” back and forth pulsing the laser for each dot.



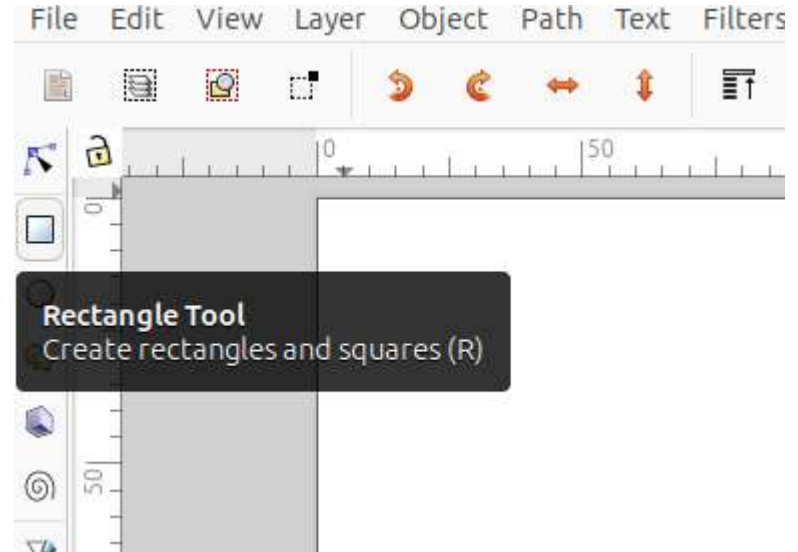
Speed and power

- Each line in your design should be given a “color”. That “color” will later determine how the cuts happen.
- Generally, each color will be assigned a “speed” and a “power” level. This will determine how the material gets cut.
- Generally, higher power = more cutting
- Lower speed = more cutting
- Etch/engrave : High speed, low power
- Cut: Low speed, high power (never above 50%)



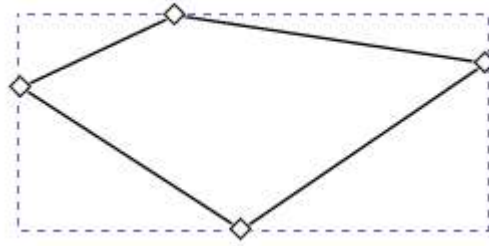
Example design

- Rectangle tool to create rectangles

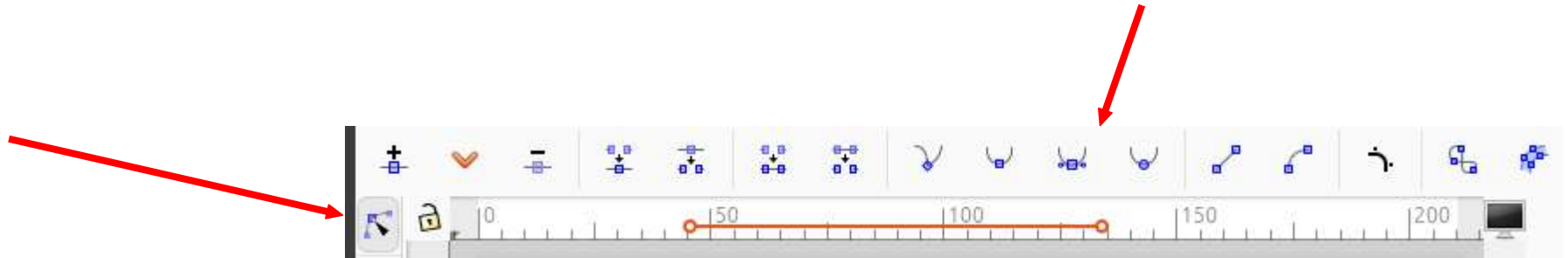


Pencil Tool

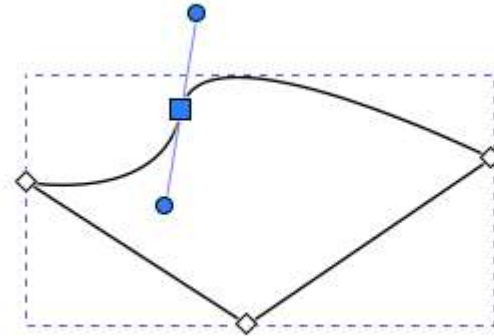
- Pencil and Text tools to straight lines
- Nodes are the endpoints
- You can drag control points to make curves



Nodes



- Adjust node control points to make curves

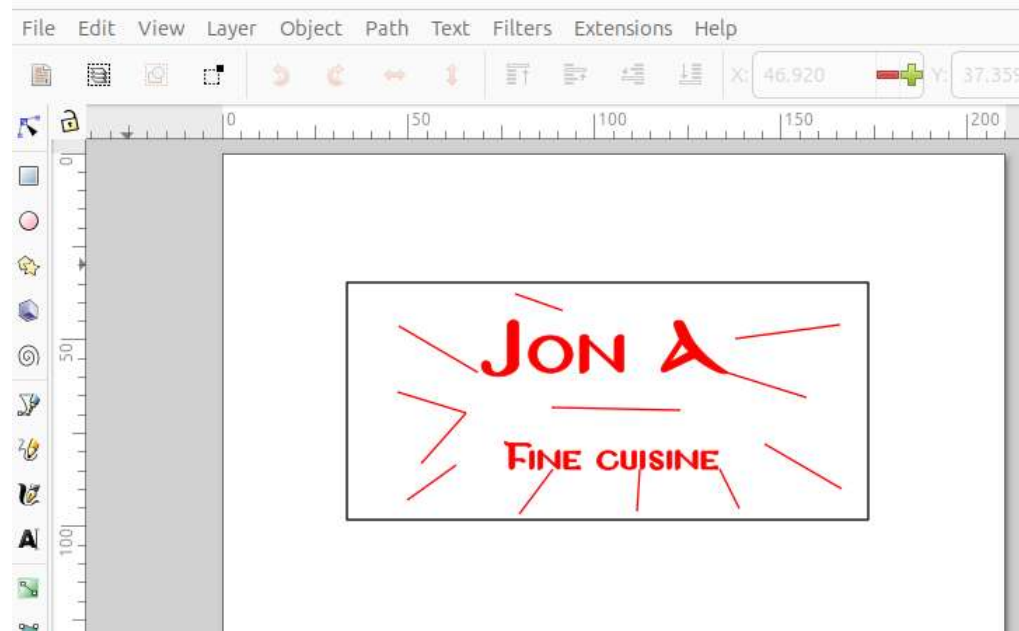


Fill and Stroke

- Set color in fill and stroke section
- Fill = None for just outline
- Fill color set for “engraving” the inside

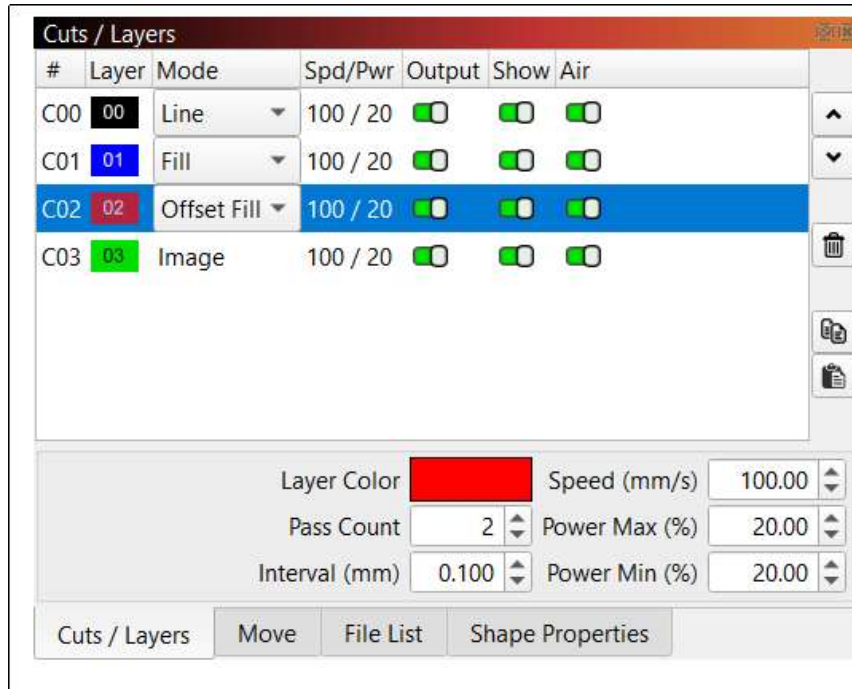


Finished Design



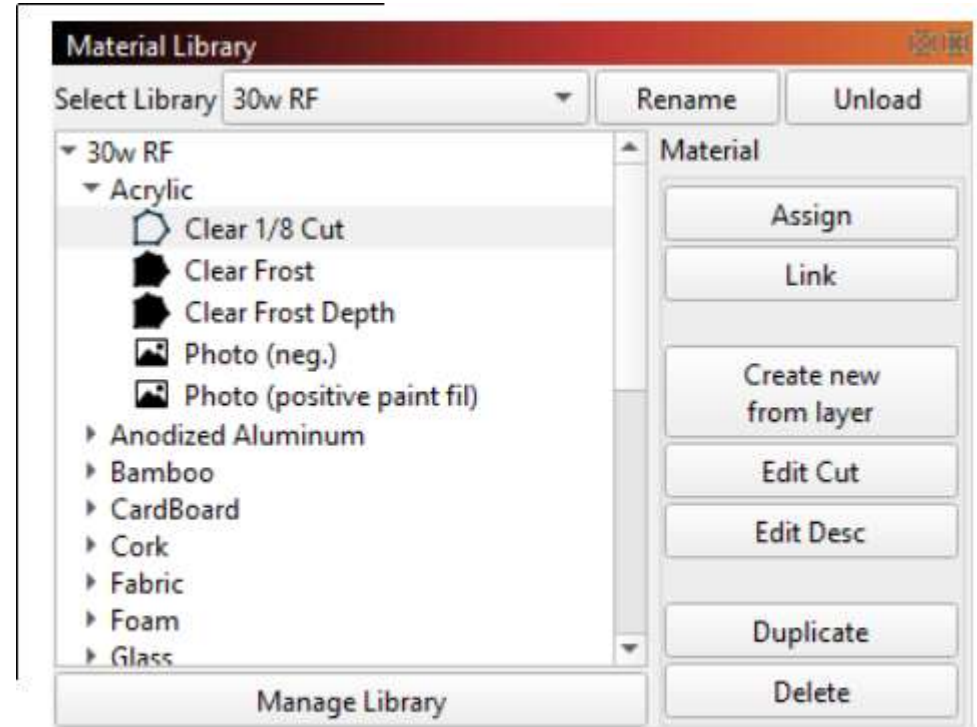
Lightburn

- Import Design
- Set speed and power settings
- Assign each color to a “mode”, and speed/power.
- Never exceed 50% power



Material Library

- You can set the speed/power manually if you want, or you can “Assign” it from the material library.
- Make sure your “Cut” layer(s) are last, otherwise your work may fall down before it does the engraving layers.



Safety

- Materials: Approved

- MDF
- Plywood and most woods
- Some plastics
- UNTREATED leather
- Stone, Glass, Ceramic

- Materials: Banned

- Plastics containing chlorine
 - Vinyl, PVC, Nylon
- Resins
- Epoxies
- Metals



Safety

- Make sure all panels are closed
- Beware of fire and flare-ups on materials. Be prepared to extinguish fire with fire blanket
- Keep lid closed to avoid blindness
- Beware in case lens vaporizes (unlikely but possible), possible Selenium exposure
- Be careful not to touch lenses or mirrors as they can bring the machine out of alignment.
- Persons under 18 must have parent or guardian nearby



Making the cuts

- Put material in machine
- Your supervisor will turn on the machine, chiller, and air assist as well as opening the air vents.
- Set focal distance between work and laser head.
- Send design to machine
- Close all doors/panels.
- Press green “Start/Pause” button.
- Don’t stare at beam
- Don’t leave unattended



Congratulations!

- You have now made your first laser-cut artifact.
- Everything else is creativity and honing your technique

